

ELWISON DENAMPO

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SKILLS

- **Backend:** TypeScript, Node.js, NestJS, Express, Prisma, Python, FastAPI, SQLAlchemy, PostgreSQL, Redis, REST APIs, JWT Auth, Socket.IO
- **Frontend:** Next.js, React, Vite, Tailwind CSS, shadcn/ui, React Query, React Hook Form, Zod, Zustand, Framer Motion, Recharts
- **DevOps & Integrations:** Docker, Turborepo, pnpm, Git, CI/CD, Vercel, Render, AWS S3, Supabase, Sentry, Stripe, DocuSign, SendGrid
- **AI/ML:** OpenAI API, Groq API, LLM Routing, Prompt Engineering
- **Other:** C#, ASP.NET Core, Entity Framework Core, SQL Server, Hangfire

WORK EXPERIENCE

JUNIOR SOFTWARE ENGINEER FutureThink Hub & La Chow **July 2025 - May 2026**

Shipped two production platforms — an AI-powered adaptive learning platform (FutureThink Edge) and a unified property management SaaS (LaChowOS) — across full-stack development, database design, third-party integrations, and DevOps.

FutureThink Edge <https://www.futurethinkedge.org>
Next.js, TypeScript, FastAPI, PostgreSQL, Redis, OpenAI API

Production EdTech platform with multi-role portals, real-time collaboration, gamification, and AI-powered adaptive learning.

- Engineered a full-stack adaptive learning platform serving 6 user roles with 86 REST API endpoints achieving P95 latency of 29ms under load — 85% below the 200ms SLA target
- Architected a config-driven multi-LLM routing layer across 4 AI providers (Gemini, DeepSeek, Grok, Claude), reducing per-query cost ~60% via subject-specific model selection and a <5ms pre-LLM curriculum enforcement classifier
- Implemented a mental-health ML ensemble (RoBERTa + VADER + TextBlob) with real-time crisis detection and 988-Lifeline escalation, achieving FERPA/GDPR/COPPA compliance across 7 data-protection endpoints with 0 high-severity CVEs
- Scaled backend to sustain 492 req/s under 500 concurrent users (k6 stress test, 590K+ iterations) and built 1,870+ automated tests (pytest, Vitest, Playwright E2E) eliminating 100% of type errors to reach Six Sigma 5.9 quality
- Designed a Bayesian Knowledge Tracing engine with VARK+ 8-dimensional learning-style profiling and 39+ cognitive training games — delivering personalized ADHD-optimized instruction across subject-isolated AI classrooms

LaChowOS <https://portal.thelachow.com/>
Next.js 16, NestJS 11, Prisma, PostgreSQL, Stripe, Turborepo, AWS S3, Redis

Production EdTech platform with multi-role portals, real-time collaboration, gamification, and AI-powered adaptive learning.

- Architected a Turborepo monorepo (pnpm workspaces) shipping a NestJS 11 API with 52 feature modules, a Next.js 16 / React 19 frontend with 62 admin + 34 tenant-portal pages, and a shared Prisma package with 84 models generating type-safe clients across both apps
- Designed PostgreSQL schema with tsrange exclusion constraints + GiST indexes preventing double-booking of shared kitchen spaces at the database level, eliminating application-layer race conditions
- Integrated 9+ third-party services: Stripe (rent + webhooks), DocuSign (eSignature lease workflows), SendGrid, AWS S3 (presigned-URL uploads + PDF parsing), QuickBooks, Supabase, Redis/ioredis caching, Sentry, and DoorLoop (idempotent legacy import via `externalId` deduplication)
- Implemented JWT authentication (Passport) with TOTP 2FA + QR enrollment, role-based access control, customizable tenant setup-link expiry (1–90 days), and proactive session-expiry warnings
- Engineered Socket.IO real-time notifications, cron-scheduled jobs (lease expiration alerts, compliance reminders), and a per-property late-fee policy engine with bulk-apply UI and read-time fee resolution

PROJECTS	BSDOC - Personal Health Management Platform https://bsdoc-project.vercel.app Next.js, Node.js, Supabase, TypeScript Health management platform with multi-role auth, admin panel, and doctor verification workflows. Built 8+ REST API routes (appointments, doctors, availability, notifications, admin analytics) with JWT authentication and Supabase Admin client, handling ~300 req/min throughput Engineered AI-powered fuzzy symptom search with 3-tier resolution pipeline (direct lookup → synonym match → corpus match) using string-similarity, reducing unresolved inputs by 78% Implemented role-based access control middleware with Supabase SSR supporting 3 user tiers (patient/doctor/admin), achieving 100% route protection with single-DB-call role verification Developed admin analytics dashboard with 5 concurrent Chart.js visualizations (bar, line, weight charts) and parallel API fetching, reducing dashboard load time by 60% Architected end-to-end appointment booking pipeline with atomic slot conflict detection (**65% faster** via <code>.limit(1)</code> optimization) and transactional push notifications (~120ms delivery) Integrated Supabase Realtime for live push notifications, eliminating polling and reducing bandwidth usage by 95%
	AI.pollo — AI Mood-Based Music Recommender https://ai-pollo.vercel.app React, TypeScript, Vite, FastAPI, PostgreSQL, Spotify API, Vercel AI-powered mood-based playlist recommender using a custom Curated Intersect Algorithm that scrapes and scores 2,000+ tracks from curated Spotify playlists against listening history and ML feedback, achieving 44% faster recommendations than deprecated endpoints. Engineered a 5-tier weighted scoring pipeline processing 2,000+ candidate tracks with <code>asyncio.gather</code> concurrency, reducing recommendation latency by 44% Built real-time multiplayer Spotify Blend rooms with 5-letter shortcode matchmaking and group consensus scoring, cutting collaborative generation time by 67% Implemented persistent ML feedback loop (thumbs up/down) biasing artist-weighting by +200 points, improving personalization hit-rate by 200% Designed dual-layer content filter (16-token blocklist + popularity gating) eliminating 95% of junk tracks from scraped playlist results Deployed serverless FastAPI backend on Vercel with stateless OAuth 2.0, automated Alembic migrations on cold boot, and async background data retention scheduler
	SchemaFlow — Interactive Schema Visualizer https://schemaflow-pi.vercel.app React 19, TypeScript, Vite, React Flow, Dagre, Zustand, Zod, CodeMirror 6, TailwindCSS Interactive schema visualization tool that parses SQL DDL and JSON payloads into live ERD diagrams with field-level edge connections, auto-layout, and relationship labels. Engineered a real-time SQL DDL and JSON schema parser processing 50-table schemas in < 2ms (~25K tables/sec), reducing manual ERD creation time by 95% using regex-based parsing and recursive descent Architected a zero-latency React Flow transformation pipeline achieving 185K transforms/sec with Dagre auto-layout, field-level edge routing, and Zod runtime validation at 100% accuracy across 5,000 iterations Built a multi-format export engine generating Mermaid ERD, Markdown docs, TypeScript interfaces, and mock REST API specs in 0.025ms per export (~40K exports/sec), enabling one-click schema documentation for Agile workflows Developed an interactive schema visualization tool with CodeMirror 6 editor, localStorage persistence, dark/light theming, and keyboard shortcuts, delivering sub-5ms end-to-end render times for complex schemas with 300+ fields
CERTIFICATIONS	SEO Certification — HubSpot Academy Valid: Dec 2025 – Jan 2027
AWARDS	2nd Prize — Huawei ICT Competition (Network Track), Philippines (2025–2026) Demonstrated knowledge in SEO fundamentals, keyword research, ranking factors, and optimization strategies.
EDUCATION	Bachelor of Science in Computer Engineering (Tertiary) August 2022 - May 2026 Cebu Institute of Technology - University (Cum Laude) Science, Technology, Engineering, and Mathematics
	Strand (Secondary) June 2016 - July 2022 University of Cebu - Pri (High Honors)